Online

Object Oriented Programming

Computer Systems Engineering

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Activity 2: Analyzing the American Stock Exchange

Module 2: Methods and Standard I/O Streams

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1. Description of the solution

This activity models a stock from the American Stock Exchange with an object-oriented approach. The program is split into two files: the **Stock** class, which holds the data and behavior of a single stock, and a test program that creates one stock and reports its price change.

The Stock class has four private data fields: **symbol** and **name** (both *String*), and **previousClosingPrice** and **currentPrice** (both *double*). The constructor receives the symbol and the name when the object is created. The two prices are assigned afterward through **setPreviousPrice** and **setNewPrice**. The method **getChangePercent** applies the formula below, so the result is positive when the price rises and negative when it falls. Each field also has a getter, so the values can be read from outside the class while the fields stay private and the object remains properly encapsulated.

```
getChangePercent = ((currentPrice − previousClosingPrice) /
previousClosingPrice) × 100
```

The test program **TestStock** creates a Stock for Netflix with the symbol NFLX and the name Netflix Corporation. It sets the previous closing price to 337.5 and the current price to 345.23, and then prints the stock data together with the price-change percentage. With those values the change is ((345.23 − 337.5) / 337.5) × 100 = **2.29%**, a small rise that the program also states in words.

2. Source code

`Stock.java`

```java
// Models a stock, storing its data and computing its price change.
public class Stock {
    private String symbol;               // the stock's symbol
    private String name;                 // the stock's name
    private double previousClosingPrice; // price for the previous day
    private double currentPrice;         // price for the current time

    // Constructor: creates a stock with the specified symbol and name.
    public Stock(String symbol, String name) {
        this.symbol = symbol;
        this.name = name;
    }

    public String getSymbol() { return symbol; }
    public String getName() { return name; }
    public double getPreviousClosingPrice() { return previousClosingPrice; }
    public double getCurrentPrice() { return currentPrice; }

    // Sets the value of the data field previousClosingPrice.
    public void setPreviousPrice(double previousClosingPrice) {
        this.previousClosingPrice = previousClosingPrice;
    }

    // Sets the value of the data field currentPrice.
    public void setNewPrice(double currentPrice) {
        this.currentPrice = currentPrice;
    }

    // Returns the percentage changed from previousClosingPrice to currentPrice.
    public double getChangePercent() {
        return ((currentPrice - previousClosingPrice) / previousClosingPrice) * 100;
    }
}
```

TestStock.java

```
// Test program for the Stock class.
public class TestStock {
    public static void main(String[] args) {
        // Create a Stock object for Netflix with its symbol and name
        Stock netflix = new Stock("NFLX", "Netflix Corporation");

        // Set the previous closing price and the new current price
        netflix.setPreviousPrice(337.5);
        netflix.setNewPrice(345.23);

        // Display the stock information
        System.out.println("Stock symbol: " + netflix.getSymbol());
        System.out.println("Stock name: " + netflix.getName());
        System.out.printf("Previous closing price: $%.2f%n",
netflix.getPreviousClosingPrice());
        System.out.printf("Current price: $%.2f%n", netflix.getCurrentPrice());

        // Display the price-change percentage
        double change = netflix.getChangePercent();
        System.out.printf("Price-change percentage: %.2f%%%n", change);

        // Indicate whether the stock went up or down
        if (change > 0) {
            System.out.println("The stock price has gone up.");
        } else if (change < 0) {
            System.out.println("The stock price has gone down.");
        } else {
            System.out.println("The stock price has not changed.");
        }
    }
}
```

3. Execution evidence

The two files compile with `javac` without errors or warnings, and the test program produces the expected output. The screenshot below shows the compilation and the run:

```
$ javac Stock.java TestStock.java
$ java TestStock
Stock symbol: NFLX
Stock name: Netflix Corporation
Previous closing price: $337.50
Current price: $345.23
Price-change percentage: 2.29%
The stock price has gone up.
```

The current price (345.23) is higher than the previous closing price (337.5), so the change is positive: the Netflix stock went up 2.29% from one day to the next.